

AI might be a net positive for the environment

Posted by u/Reynvald • 0 points • 21 comments

Please take all calculations and sources with a grain of salt, as such things are generally hard to quantify. I also would be happy to get corrected if I made mistakes or misrepresented some data. I tried to balance myself out with sources from both sides. I'm an AI doomer myself, but at the same time I think that many environmental claims is an overreach. So I'm obviously have some biases.

IMO there are three main talking points about AI, harming the environment:

- Energy consumption
- Carbon footprint and Greenhouse gases (GHG) in general
- Water scarcity and pollution

I sorted sources about negative impacts at the top of each section and about negative impact - on the bottom.

1. Energy consumption

As of 2024, [Data centers accounted for about 1.5% of global electricity consumption, with AI accounted for 15% of total data centre energy demand accordingly.](<https://www.scientificamerican.com/article/ai-will-drive-doubling-of-data-center-energy-demand-by-2030/>) Therefore we can say that AI itself is using around 0.225% of global energy reserves.

[Predicted share of energy usage for data centers by 2030 is between 5 and 20%.]

(<https://www.scientificamerican.com/article/ai-will-drive-doubling-of-data-center-energy-demand-by-2030/>)

Considering that AI it still on it's growth and can take over up to 50% of all data center's resources, in 2030 it can be responsible for 2.5 up to 10% of all energy consumption (20 up to 90 times more, than of now) which is quite radical, but not unrealistic prediction.

Nevertheless, as of right now, [ML-related technologies is able to provide 15% improvement in grid efficiency and 10–20% increase in battery storage efficiency](<https://www.mdpi.com/2073-4433/15/10/1250>) and [20–30% relative efficiency gains in cell and module R&D.]

(<https://journals.sagepub.com/doi/full/10.1177/0958305X241256293>)

Same magnitude of efficiency gains is also the case for all clean and non-clean energy sources, by forecasting the weather and autoadjusting solar panels, micromanaging power grids and plants, predicting deposits of fossil energy sources and so on.

I think safe to say, that estimated energy gain overall will equal to or most likely surpass even the most pessimistic prognosis of 10% energy consumption from AI alone by 2030.

2. Carbon footprint and GHG in general

According to [ICEF report from November 2024](https://www.icef.go.jp/wp-content/themes/icef/pdf/2024/roadmap/15_ICEF2.0%20GHG%20Emissions%20from%20AI_stand%20alone.pdf), (This link will download PDF file!) AI's total GHG emissions are estimated at 100–300 million tonnes CO₂, or roughly 0.2-0.6% of global emissions. With that, operational emissions are around 0.05% while manufacturing servers, chips, facilities, model trainings and life-cycle impacts make up the remainder.

At the same time AI can [reduce global GHG emissions by 5–10% by 2030] (<https://thesustainableagency.com/blog/environmental-impact-of-generative-ai/>), via optimized grids, predictive maintenance, and smart agriculture and, additionally, [cuts of up to 5.3 gigatons CO₂ (another 5–10% of current emissions)](<https://www.sciencedirect.com/science/article/pii/S277242712200122X>) - through applications in transport, buildings, and supply chains.

[One specific research](<https://www.nature.com/articles/s41598-025-97110-3>) (from month ago) from China indicates, that correlation between % of AI adoption and % of reducing carbon footprint (1% and 0.0395% accordingly) is quite sustainable and universal across the industries. But hard to say of this correlation will hold with future increasing Alfication of industries.

3. Water scarcity and pollution

[Apparently in US AI is responsible for 0.5-0.7% of total annual water withdrawal] (<https://oecd.ai/en/work/how-much-water-does-ai-consume>). If source took a data of water consumptions by data centers in general (it most likely the case, as most of the articles do so), then actual numbers will be a 15% of 0.5-0.7%, which is 0.075-0.105% accordingly.

Considering that most of the world AI infrastructure is located in US and China, safe to say, that for the rest of the world this percentages is significantly smaller.

The real concern, however, is the water pollution and separate cases of mismanagement from the corporations. [Quote: "Google's planned data centre in Uruguay, which recently suffered its worst drought in 74 years, would require 7.6 million litres per day, sparking widespread protest."] (https://sdgs.un.org/sites/default/files/2024-05/Gupta%2C%20et%20al._AIs%20excessive%20water%20consumption.pdf) (This link will download PDF file!)

[Recent article from Politico](<https://www.politico.com/news/2025/05/06/elon-musk-xai-memphis-gas-turbines-air-pollution-permits-00317582>) about air pollution from xAI data center is also seems to me as a fair critique.

Here is a positive aspects:

[AI irrigation can reduce water usage by 30-50% while increasing yields by 20–30% (which is 5–8%

savings of global agricultural withdrawals if deployed worldwide).]
(<https://www.sciencedirect.com/science/article/pii/S2772375525002151>)

[AI acoustic and pressure-based leak detection is already working and have 80–97% accuracy, cutting non-revenue water losses by 20–40%. Given that networks lose ~30% of supply globally (the most distant and arid places usually suffer the most), AI is saving 6–12% of treated water.

](https://www.waterrf.org/serve-file/2024_IWS-Challenge-Solution_Case-Western-CWD.pdf)

(This link will download PDF file!)

Same goes for demand forecasting, pump optimization, water quality assessment and many other projects, totaling up to 12% of the saved fresh water worldwide (if implemented worldwide as well, which is not the case for now). Some of this solutions is already implemented and working, although mostly in the water hungriest areas, like parts of Africa, China and India.

I think it's crucial to point out, that most of the water scarcity-related suffering is mostly occurring far from data centers and their water sources. And this problem is more of a logistical one (how to transport the water to the arid areas), than of sheer amount of fresh water world supplies.

I want to highlight, that AI still have an impact on environment and it's a right thing to strife for reducing the environmental impact in any area. But I believe that misinformation, toxicity and alarmism eventually will harm the both sides of this debates.

Comments

u/[deleted] • 1 points • 2026-04-26 14:43 UTC

[removed]

u/Mohammad_Nasim • 2 points • 2025-07-25 15:51 UTC

I really like how this discussion highlights AI's positive impact on the environment. I've been using Kumo by SoranoAI, an AI agent that provides clear weather insights just by asking in natural language no coding or heavy computing needed. It's a great example of AI tools that are both smart and eco-friendly.

u/Dazzling_Occasion_47 • 1 points • 2025-05-14 20:08 UTC

Regarding the percentage of energy consumption from AI data centers, there's one thing you're missing. It's a marginal increase in energy consumption, but the margins are what matter actually.

At the moment, despite all the efforts, investments and improvments in wind, solar, storage, blah blah blah, we, as in *the world*, are making zero progress in reducing CO2 emissions. Why? Because our marginal increase in global energy consumption outweighs the net rate at which carbon-based energy generation resouces are being replaced by carbon free ones. As long as this trend continues, we will never solve the climate crisis, and anything that adds to that marginal increase is counter-productive.

The future, of course, is unpredictable, and how AI may be used to solve various engineering problems isn't something we can predict, so perhaps there is promise in the future, perhaps AI will render humans obsolete, and since humans are the biggest carbon-intensive resource, this could be positive... but at the moment AI is just yet another energy consumer added to the grid, and it's not presently accomplishing anything other than making it easier for me to google.

u/Reynvald • 1 points • 2025-05-14 20:46 UTC

Thanks for the detailed reply.

>At the moment, despite all the efforts, investments and improvements in wind, solar, storage, blah blah blah, we, as in *the world*, are making zero progress in reducing CO2 emissions.

I would say that progress is here, including the positive temp of growth of renewable energy share. 50%+ for the last two years, actually. And slight reduce of fossil fuel share. But yeah, global demand on energy is rising as well as sheer number of energy and carbon footprint. No argue here.

But still, I think it's important, that from all other industries, that can, at best, reduce their impact as much as possible (but never to zero, just due to the nature of this industries), AI has a real possibility to be energy/carbon negative (in a good sense) in a next decade. So I think we should push governments to control AI in this matter, sure. As any other industry. But not to the detriment and postponement of it's environmentally beneficial projects.

>but at the moment AI is just yet another energy consumer added to the grid, and it's not presently accomplishing anything other than making it easier for me to google.

I can send tonn of links with already implemented and cool use cases in medicine, science, production, but since we talking on a global scale, this individual projects would be a bad faith argument from my side.

Just will point out that as of now, AI, presumably, add the equivalent of \$2.6 trillion to \$4.4 trillion annually GDP worldwide. Around GDP of the entire Great Britain.

<https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/the-economic-potential-of-generative-ai-the-next-productivity-frontier>

Sure, it's also increases consumption, but all the attempts to improve environment through reducing consumption just never worked and likely never will, leaving us with only option to consume even more and through that use additional resources on eco-positive projects. Even for me, on an individual scale, LLMs is a big help if studies and work.

u/captdunsel721 • 1 points • 2025-05-12 11:56 UTC

You've done the research on the information currently available. Might be a net positive. Might also contribute to fossil fuel growth or justification for its continued use. Which could result in increased climate chaos, mass human migration, longer more destructive wars. But less human life- might be a net benefit for the planet. You're blindly feeling one end of a very large, rapidly changing mammal... careful where you stick your fingers.

u/Reynvald • 1 points • 2025-05-12 12:12 UTC

I just wanted to stick my fingers to somewhere :(

On a serious note, I find fossil fuel argument plausible but not to such dire degree. And this is not unique problem for this field. It's generally more up to the state and will of the politicians. The same way, the AI industry can give a push to the development of nuclear power plants.

I try to learn more about this topic exactly to not touching elephant blindly.

u/captdunsel721 • 1 points • 2025-05-14 17:07 UTC

Late response to the "not to such a dire degree" I guess it depends on location, location, location.

Excerpt from a Heather Cox Richardson posting dated May 13th.

"Musk powers his AI from a massive supercomputer in Memphis, Tennessee. As Dara Kerr of The Guardian reported last month, the Southern Environmental Law Center discovered that Musk had quietly moved at least 35 methane-powered generators—enough to power a city—to the plant to help power the supercomputer he calls "Colossus," which powers his chatbot "Grok." Those generators are unpermitted and are major producers of carcinogens and other toxins. After the company assured Memphis mayor Paul Young that only 15 of the generators were on, thermal imaging showed at least 33 running. The supercomputer is in a historically Black neighborhood with a history of industrial pollution and higher rates of cancer and asthma than other Memphis neighborhoods. When residents

spoke out against the supercomputer, a group calling itself "Facts Over Fiction" but without any other identifying information spread flyers claiming the turbines are "specially designed to protect the air we all breathe." They also claimed that the Environmental Protection Agency and the county health department regulate the generators, but both agencies told Kerr that they had not issued permits for their use at the Memphis plant. In March, Musk bought another property in Memphis to expand the plant by a million square feet."

u/Reynvald • 1 points • 2025-05-14 17:12 UTC

Yes, it surely depends on location and many socio-ecological factors. But due to how you formulated your first comment:

>increased climate chaos, mass human migration, longer more destructive wars. But less human life- might be a net benefit for the planet.

I assumed that you talking about global or near global scale. And respond accordingly.

u/captdunsel721 • 1 points • 2025-05-15 02:45 UTC

A little AI here, a little AI there, some in Texas, some in CA. China wants some, Russia wants some, etc. Not long before local elections becomes global. If AI was the only fossil fuel sucking game in town, or if it were powered with renewables, maybe it could be a net positive. But, alas, it will end up being a dire degree... and your generation will be wondering what the hell hit it.

u/Reynvald • 1 points • 2025-05-15 02:49 UTC

I'm pretty sure I will know what was hit as. I get your point, but we just have a different view on thigs, apparently.

u/captdunsel721 • 1 points
• 2025-05-15 03:09 UTC

Yes, we do. Have a good night.

u/[deleted] • 1 points • 2025-05-12 06:56 UTC

[removed]

u/Velocipedique • 1 points • 2025-05-11 23:50 UTC

Water-wise, anticipated AI centers in the Austin TX area will be the last straw, so to speak, in the limited supply of Lake Travis and the Rio Colorado. When it comes to droughts mother nature will always win!

u/Reynvald • 1 points • 2025-05-12 06:53 UTC

Yea, huge additional water consumption in already arid regions is a legitimate concern.